Serie 5

Aufgabe 1

Gleichung: $\exp(x^2) + x^{-3} - 10$

3 Lösungen in $x \in [-3, 3$

2. $e^{x^2} + x^{-3} - 10 \rightarrow e^{x^2} + \frac{1}{x^3} - 10$

Intervalle mit Nullstellen, heisst: $f(a) \cdot f(b) < 0$ aber $x \neq 0$
→ sonst Division durch 0

wenn $x = -1 = f(-1) = e^{-1^2} + \frac{1}{(-1)^3} - 10 = e^1 + \frac{1}{-1} - 10 \approx -8.28$
$x = -2 = f(-2) = e^{-2^2} + \frac{1}{(-2)^3} - 10 = e^4 + \frac{1}{-8} - 10 \approx 44.47$
=> Vorzeichenwechsel!

Eine Nullstelle liegt im Intervall $[-2, -1]$.

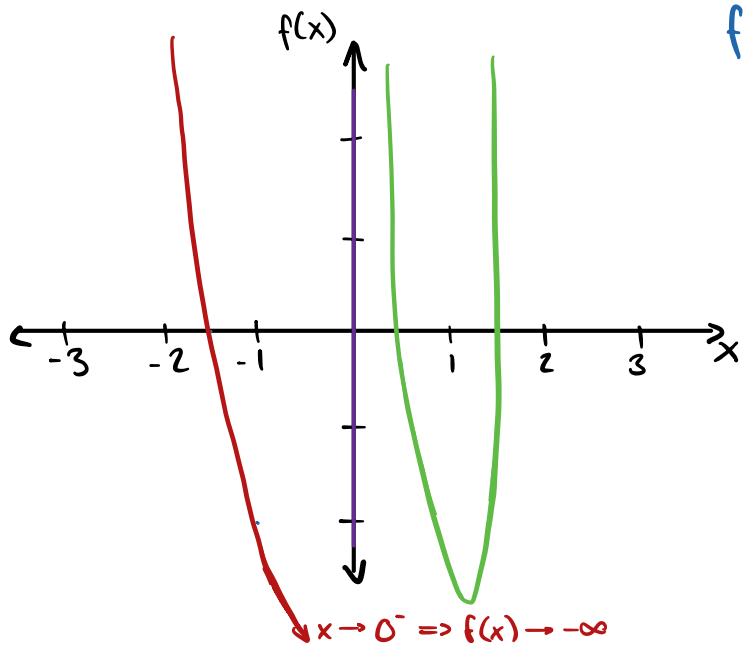
$x = -3 \rightarrow e^{-3^2} + \frac{1}{-3^3} - 10 = e^9 + \frac{1}{-27} - 10 \approx 8093.05 \rightarrow$ positiv !!
$x = 1 \rightarrow e^{1^2} + \frac{1}{1^3} - 10 = e^1 + \frac{1}{1} - 10 \approx -6.28 \rightarrow$ negativ!!
$x = 2 \rightarrow e^{2^2} + \frac{1}{2^3} - 10 = e^4 + \frac{1}{8} - 10 \approx 44.72 \rightarrow$ positiv!!
$x = 3 \rightarrow e^{3^2} + \frac{1}{3^3} - 10 = e^9 + \frac{1}{27} - 10 \approx 8093.12 \rightarrow$ positiv!!

$x = 0.25 \rightarrow e^{0.25^2} + \frac{1}{0.25^3} - 10 \approx 55.065 \rightarrow$ positiv!
$x = 0.5 \rightarrow e^{0.5^2} + \frac{1}{0.5^3} - 10 \approx -0.716 \rightarrow$ negativ!

Nullstellen liegen in den Intervallen

$[-2, -1]$, $[0.25, 0.5]$, $[1, 2]$

Funktionsskizze:



$f(x) = e^{x^2} + \frac{1}{x^3} - 10$

$f(x) \{-3 \leq x < 0\}$

$f(x) \{0 < x \leq 3\}$

$f(x) \{x = 0\}$

Newton-Verfahren:

Iterationsformel: $x_n = x_{n-1} - \frac{f(x_{n-1})}{f'(x_{n-1})}$

$f'(x) = 2xe^{x^2} - \frac{3}{x^4}$

$x_0 = 2 \Rightarrow x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$

$\Rightarrow x_1 = 2 - \frac{44.72}{218.205} \approx 1.795$

$x_2 = 1.795 - \frac{f(1.795)}{f'(1.795)} \Rightarrow 1.795 - \frac{15.252}{89.744} \approx 1.6250$

$x_3 = 1.6250 - \frac{f(1.6250)}{f'(1.6250)} \Rightarrow 1.6250 - \frac{4.2550}{45.1411} \approx 1.5307$

$x_4 = 1.5307 - \frac{f(1.5307)}{f'(1.5307)} \Rightarrow 1.5307 - \frac{0.6917}{31.3315} \approx 1.509$

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vereinfachtes Newton-Verfahren:

$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_0)}$

$f(x) = e^{x^2} + \frac{1}{x^3} - 10$

$f'(x) = 2xe^{x^2} - \frac{3}{x^4}$

$x_0 = 0.5$

$f'(x_0) = f'(0.5) \approx -46.716$

$x_1 = 0.5 - \frac{f(0.5)}{-46.716} = 0.5 - \frac{-0.716}{-46.716} \approx 0.4847$

$x_2 = 0.4847 - \frac{f(0.4847)}{-46.716} = 0.4847 - \frac{0.0466}{-46.716} \approx 0.4857$

$x_3 = 0.4857 - \frac{f(0.4857)}{-46.716} = 0.4857 - \frac{-0.0063}{-46.716} \approx 0.4855$

$x_4 = 0.4855 - \frac{f(0.4855)}{-46.716} = 0.4855 - \frac{0.00422}{-46.716} \approx 0.4856$

Sekantenverfahren

$x_{n+1} = x_n - \frac{f(x_n)(x_n - x_{n-1})}{f(x_n) - f(x_{n-1})}$

Startwerte: $x_0 = -1.0$

$x_1 = -1.2$

$f(x) = e^{x^2} + \frac{1}{x^3} - 10$

$x_2 = x_1 - \frac{f(x_1)(-1.2 - (-1.0))}{f(x_1) - f(x_0)}$

$x_2 = -1.2 - \frac{f(-1.2)(-1.2 + 1.0)}{f(-1.2) - f(-1.0)} \Rightarrow -1.2 - \frac{(-6.3580) \cdot (-0.2)}{(-6.3580) - (-8.2817)} = -1.2 - \frac{1.2716}{1.9237} \approx -1.860$

-1.01673

$$x_3 = -1.8610 - \frac{f(-1.8610)(-1.8610+1.2)}{f(-1.8610)-f(-1.2)} = -1.8610 - \frac{21.7677\cdot(-0.661)}{21.7677-(-6.358)} = -1.861 - \frac{-14.3885}{28.1257}$$

$x_3 \approx -1.3494$

$$x_4 = -1.3494 - \frac{f(-1.3494)(-1.3494+1.8610)}{f(-1.3494)-f(-1.8610)} = 1.3494 - \frac{(-4.2294)\cdot 0.5116}{(-4.2294)-(21.7677)} = -1.3494 - \frac{-2.1638}{-25.9971}$$

$x_4 \approx -1.4326$

$$x_5 = -1.4326 - \frac{f(-1.4326)(-1.4326-(-1.3494))}{f(-1.4326)-f(-1.3494)} = -1.4326 - \frac{-2.5540\cdot(-0.0832)}{-2.5540-(-4.2294)}$$

$$= -1.4326 - \frac{0.2125}{1.6756} = x_5 \approx -1.5594$$